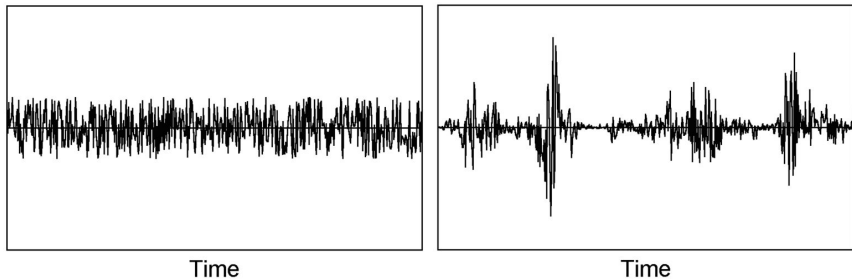


Wavelet ARIMA-GARCH and Long-Trend Portfolios

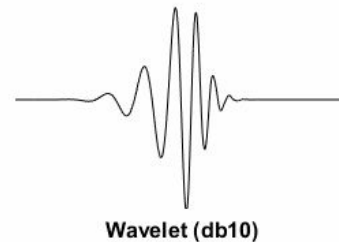
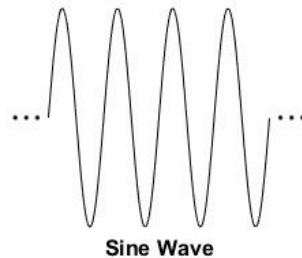
BQT Project Presentation

What are Wavelets?



- Wavelets have a more local structure
- Better fit for reconstructing a heteroskedastic time series by frequency levels

- Equity time series data are heteroskedastic
- We cannot fit a single frequency component across the horizon



Our Past Results

- Utilized DWT (Discrete Wavelet Transformation) to preprocess our data (decomposing the signal into frequency components, separating long term trends from short term noise)
- Applied AR (Auto Regressive) and MA (Moving Average) to predict the expected value at specific point in time based on previous errors

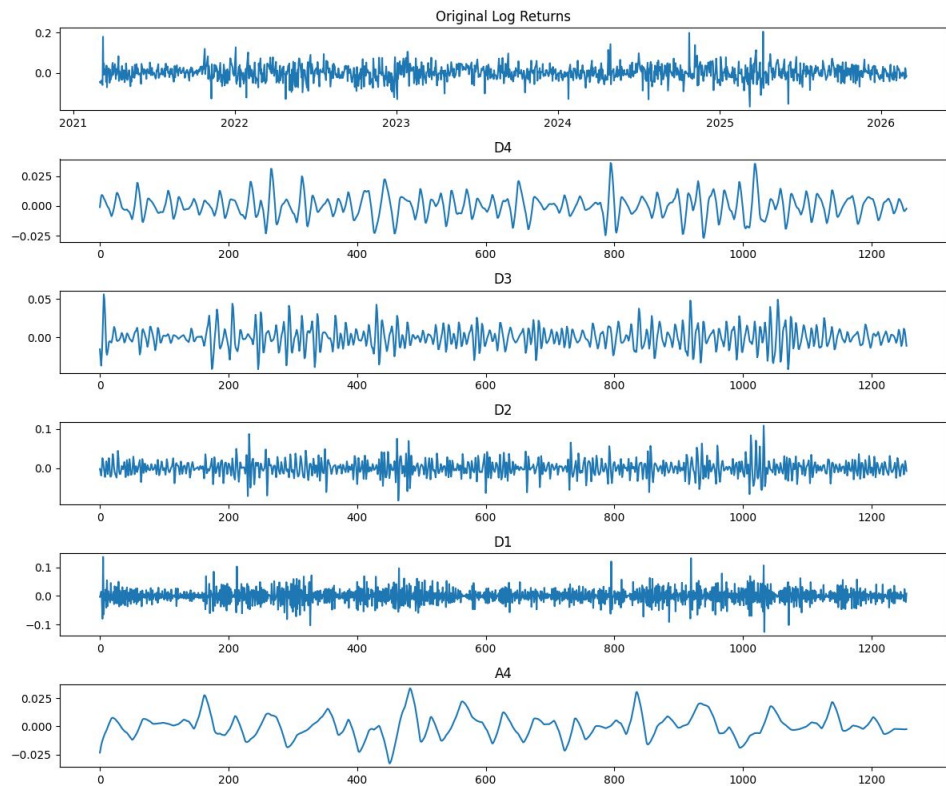
$$X_t = c + \phi_1 X_{t-1} + \dots \phi_p X_{t-p} + \epsilon_t \quad X_t = \mu + \epsilon_t + \theta_1 \epsilon_{t-1} + \dots + \theta_q \epsilon_{t-q}$$

- Applied GARCH (Generalized Autoregressive Conditional Heteroskedasticity) to define variance of the error as an equation of past error

$$\sigma_t^2 = \omega + \alpha_1 \epsilon_{t-1}^2 + \dots + \alpha_q \epsilon_{t-q}^2$$

- The result: The ARIMA-GARCH model is a commonly used forecasting model to identify linear trends changing levels of volatility over time
- Our Goal: Combine wavelet transformation with the ARIMA-GARCH frameworks to develop a model that performs better than the standard ARIMA-GARCH model

Wavelet Decomposition



	Energy	Percent Contribution
A4	0.122602	6.904772
D1	0.908668	51.174746
D2	0.446922	25.169910
D3	0.197918	11.146442
D4	0.099508	5.604129

⇐ Wavelet Decomposition

Past Results cont.

ARIMA-GARCH:

- Hovers around 0
- Less capable of separating noise from trend

Wavelet-ARIMA-GARCH:

- More meaningful trends
- Filters through high-frequency noise

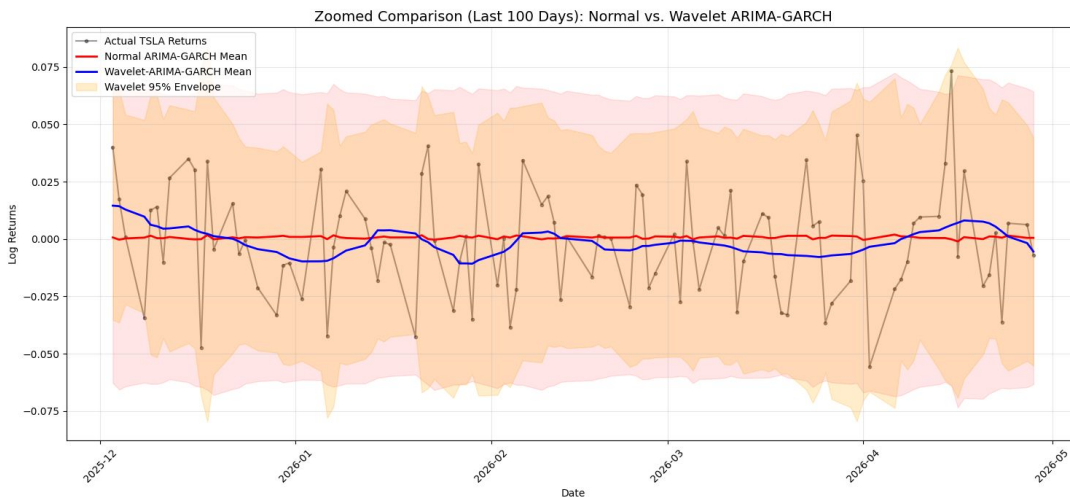
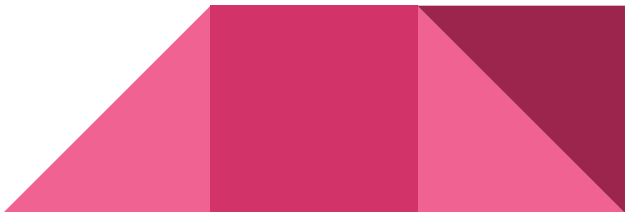


Figure. Forecast mean values from backtesting

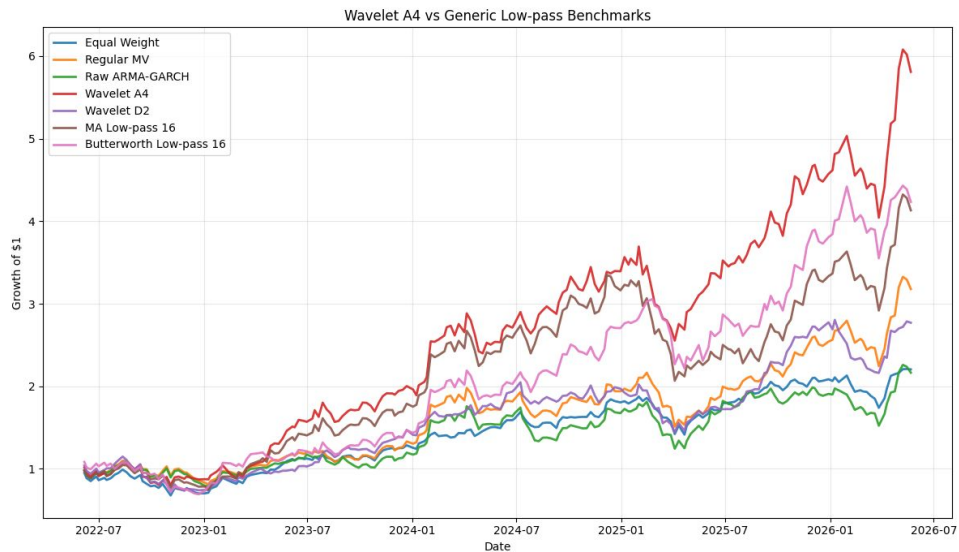
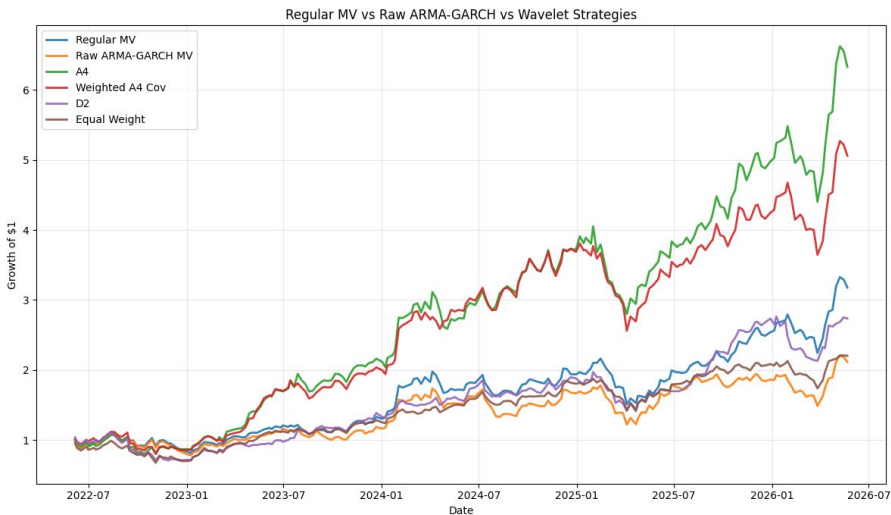
Our Extensions: Building a Mean-Variance Portfolio

- Traditional mean-variance portfolio requires forecast values of mean and variance
- We use the improved forecast values from wavelet-ARIMA-GARCH to build a better portfolio

$$\min_w w^T \Sigma w - q R^T w$$

- w : Vector of portfolio weights (decision variable)
 - Σ : Covariance matrix of asset returns (measures risk)
 - R : Vector of expected returns for each asset (measures return)
 - q : Risk aversion parameter (higher q means more return-focused)
 - $w^T \Sigma w$: Portfolio variance (total risk)
 - $R^T w$: Portfolio expected return
- 

Portfolio Performance



Summary

- Using A4 level for portfolio works like a low-pass filter, but it is more robust due to the low frequency trend it captures.
- We ran the backtest on five big tech companies; we can diversify the portfolio by more industries.

